GARUD

Project is a cognitive surveillance system that utilizes AI to give real-time analytics & perform large no. of operations, simultaneously. The entire system will rely on localised data center with very large computing power, cloud storage for this system as well as 3rd party clients later. The time scale for this product is 4 months, but product has to be operational ready by April 30, 2026.

MVP

MVP will run on a laptop utilising the system GPU & edge chip onboard, on the MVP camera module.

Camera module while have optical sensor, lens, edge AI chip, internal on onboard SSD storage, which will be integrated with software onboard the laptop.

Software operational capabilities of MVP will be following.

1. Object detection & tracking: The system will be required to detect large number of different daily objects utilising trained model through open-source data sets, track them across the field of view, and tag them appropriately. This feature will help to identify and locate contrabands.
2. Anomaly detection: Accidents or mishaps such as crime, mob fury, attacks using weapons, or any threat to human, or animal life will be detected in real time and according to the level of intensity of the event, appropriate measures will be taken in real time and concerned authorities will be alerted. This will allow for more safety of the people and property.
3. Gesture interpretation: Littering activities in public spaces along with other civic misbehaviour such as spitting, loitering in spaces prohibited, criminal actions will be identified immediately, and

necessary actions will be taken to discourage such anti-social behaviour.

4. Crowd count: This feature will allow to maintain civil harmony and recognise patterns as a measure of preventing misdoings from happening.
5. Facial recognition: Through API integration necessary actions can be taken by identifying individuals at fault of breaking the law. For the MVP, facial data of a small test group will be enough to demonstrate the capabilities of the system.